

be overridden. These tags are attached to the header of the message streamed by the forwarder, and the original log entry from the source is available in the payload of the message packet.

Filter: This is the workhorse of the entire LMS and can be used to apply different filters to parse log entries from different sources. This helps in dealing with the discrepancy in the formats of the log entries from different sources to normalise the log entries streamed into Logstash. The type attribute added by the forwarder in the header can be used to differentiate log entries from various sources, and helps in applying appropriate filters to the log entries.

Grok is used by the Logstash filter for matching patterns of interest with the incoming log entries from the forwarder. Patterns are typically specified using regular expressions (*regex*) and for intricate patterns, specifying a *regex* would be tricky; hence, Logstash provides a *Regex* library that hosts human-readable aliases for commonly used *regex* patterns. The format for specifying patterns in Grok is *%[Syntax specification (or) pattern: Semantic actions]*. Standard *regex* patterns are available as a Logstash plugin.

Output: The output from Logstash can be used for multiple purposes like storing, notifying, relaying to another logging system, collecting metrics and management reporting. The output can be redirected to one of the 50+ destinations supported by Logstash like Elasticsearch, etc.

There are plugins available in Logstash for input, filter and output phases, and a user can write plugins for these phases to be used with Logstash too. In case of failures like the non-availability of a suitable parser script to parse the incoming log entry, non-responsive input/output sources, etc., the pipeline halts. Logstash can be downloaded from <https://www.elastic.co/downloads/logstash>.

Elasticsearch

Elasticsearch is a search and analytics tool that can be used with Logstash to find patterns of interest in the log entries in real-time (as the logs roll) and hence can provide useful real-time insights. The Elasticsearch tool can be downloaded from <https://www.elastic.co/downloads/elasticsearch>.

Kibana

Kibana is a tool that helps in visualising log reports as charts, maps, tables, etc. And it also has provisions for performing advanced data analytics and creating customisable dashboards. Kibana can be downloaded from <https://www.elastic.co/downloads/kibana>.

Some troubleshooting tips

(i) While working with the RADIUS logs, we couldn't find any Grok pattern. So we had to formulate Grok patterns for filtering from a file containing the RADIUS log. When we ran

Logstash, it did not receive any data from the file. This was because there were some errors in the Grok pattern. When we corrected these errors and re-ran Logstash, it still did not pull any data from the file. This was because Logstash keeps track of a file's last modified timestamp, and it applies the filter only when the last modified timestamp is later than the latest timestamp at which Logstash pulled in the data from the file.

(ii) Files sent by the Logstash forwarder from one system were not received by the Logstash pipeline which was running in another system. We later found that the problem was with the ssh setting. Check your ssh settings and ssh keys before running the Logstash forwarder.

Benefits

After setting up the LMS at the TCE data centre, we realised the following benefits:

- Convenience in managing and archiving logs.
- Reduction in mean time to repair (MTTR) as the root cause for the failures can be quickly analysed.
- Convenience in analysing and searching real-time rolling logs generated by transactional applications running in TCE's data centre.
- Centralised LMS set-up helps researchers and faculty of TCE in archiving log entries and creating a test set for evaluating their data analytic algorithms. 

References

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- [9] RADIUS Server - <http://en.wikipedia.org/wiki/RADIUS> [Accessed on 11-June-2015]

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